

freqspace

Frequency spacing for frequency response

Syntax

```
[f1,f2] = freqspace(n)
[f1,f2] = freqspace([m n])
[x,y] = freqspace(__,'meshgrid')
f = freqspace(N)
f = freqspace(N,'whole')
```

Description

`[f1,f2] = freqspace(n)` returns the two-dimensional frequency vectors `f1` and `f2` for an `n`-by-`n` matrix. [example](#)

`freqspace` returns the implied frequency range for equally spaced frequency responses. `freqspace` is useful when creating desired frequency responses for various one- and two-dimensional applications.

`[f1,f2] = freqspace([m n])` returns the two-dimensional frequency vectors `f1` and `f2` for an `m`-by-`n` matrix.

`[x,y] = freqspace(__,'meshgrid')` is equivalent to

```
[f1,f2] = freqspace(...);
[x,y] = meshgrid(f1,f2);
```

`f = freqspace(N)` returns the one-dimensional frequency vector `f` assuming `N` evenly spaced points around the unit circle.

`f = freqspace(N,'whole')` returns `N` evenly spaced points around the whole unit circle. In this case, `f` is $0:2/N:2*(N-1)/N$.

Examples

[collapse all](#)

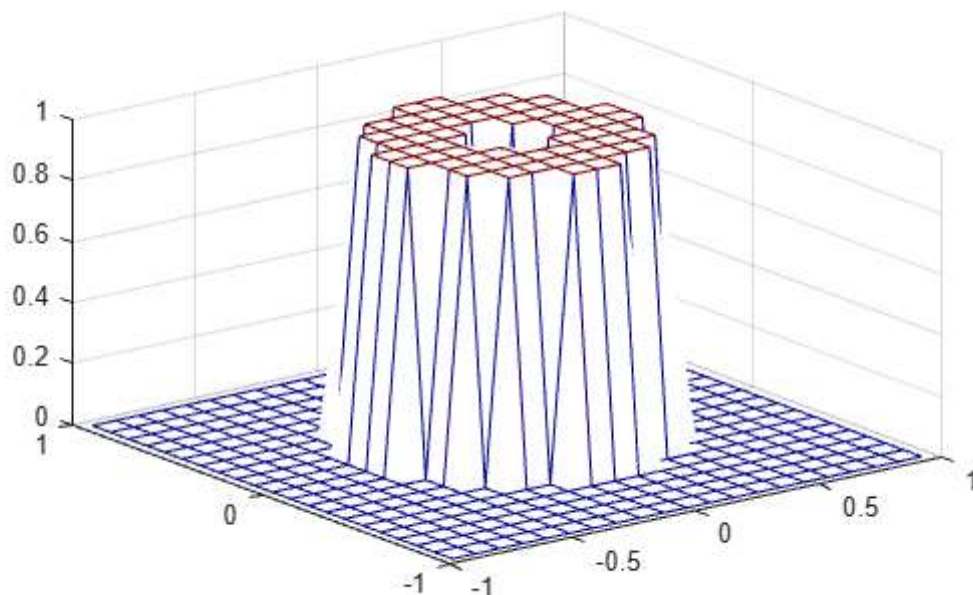
Create Matrix of Desired Passband Response Using Frequency Vectors

Create a matrix `Hd` that contains the desired bandpass response for an approximately symmetric, two-dimensional bandpass filter with passband between 0.1 and 0.5 (normalized frequency, where 1.0 corresponds to half the sampling frequency, or π radians.) Use `freqspace` to create the frequency vectors `f1` and `f2`.

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```
[f1,f2] = freqspace(21,'meshgrid');
Hd = ones(21);
r = sqrt(f1.^2 + f2.^2);
Hd((r<0.1)|(r>0.5)) = 0;
colormap(jet(64))
mesh(f1,f2,Hd)
```

[Get](#) ▼



Input Arguments

[collapse all](#)

n — Length of frequency vectors
numeric scalar

Length of the frequency vectors, specified as a positive numeric scalar.

Data Types: single | double

[m n] — Length of each frequency vector
two-element vector of positive numeric values

Length of each frequency vector, specified as a two-element vector of the form [m n].

Data Types: single | double

N — Number of points around the unit circle
numeric scalar

Number of points around the unit circle, specified as a positive numeric scalar.

Data Types: single | double

Output Arguments

[collapse all](#)

f1 — 2-D frequency vector
numeric vector

2-D frequency vector, returned as a numeric vector.

For n odd, both f_1 and f_2 are $[-n+1:2:n-1]/n$.

For n even, both f_1 and f_2 are $[-n:2:n-2]/n$.

f_2 — 2-D frequency vector

numeric vector

2-D frequency vector, returned as a numeric vector. f_2 has the same odd and even behavior as [f1](#).

x — X-coordinates over grid

numeric matrix

X-coordinates over a grid, returned as a numeric matrix.

y — Y-coordinates over grid

numeric matrix

Y-coordinates over a grid, returned as a numeric matrix.

f — 1-D frequency vector of points around the unit circle

numeric vector

1-D frequency vector of points around the unit circle, returned as a numeric vector of N evenly spaced points.

- For N even or odd, f is $(0:2/N:1)$.
- For N even, `freqspace` returns $(N+2)/2$ points.
- For N odd, `freqspace` returns $(N+1)/2$ points.

Extended Capabilities

[expand all](#)

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Thread-Based Environment

Run code in the background using MATLAB® `backgroundPool` or accelerate code with Parallel Computing Toolbox™ `ThreadPool`.

GPU Arrays

Accelerate code by running on a graphics processing unit (GPU) using Parallel Computing Toolbox™.

Version History

Introduced before R2006a

See Also

[meshgrid](#) | [fsamp2](#) (Image Processing Toolbox) | [fwind1](#) (Image Processing Toolbox) | [fwind2](#) (Image Processing Toolbox)

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